A Comparison of Dimensionality Reduction and Feature Selection for Network Intrusion Detection on Apache Spark ML with a Leakage-Aware Evaluation Pipeline

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***Abstract*—Machine-learning-based network intrusion detection systems (IDS) typically process hundreds of network-flow features, which is computationally expensive and hard to deploy at the edge. This paper presents a systematic comparative study of four dimensionality-reduction strategies on Apache Spark ML: a baseline using all features, selection by Random Forest Importance, selection by SHAP, and PCA. Each strategy is evaluated across eight classification algorithms together with two ensemble methods on the CICIDS2017 dataset. Unlike most prior work, we emphasize the *rigor of the evaluation pipeline*: (i) we remove the label-leaking feature `destination_port` and deduplicate flows at the feature level before splitting the data; (ii) we fix the comparison configuration *a priori* to avoid selection bias; (iii) we select models on a validation set held out from the training set, together with an in-distribution stability check and a *cross-dataset* generalization evaluation (between CICIDS2017 and CSE-CIC-IDS2018); (iv) we assess statistical significance using a paired permutation test over multiple data resamplings, together with Nadeau–Bengio corrected confidence intervals and effect sizes; and (v) we add a per-attack-type multiclass evaluation. The study aims to verify whether feature selection retains most of the classification performance while cutting the number of features by about 60%, and whether SHAP Top-30 balances accuracy, explainability, and cost so as to be suitable for export to the Jetson Orin Nano Super edge device (deployed in a companion systems paper); the quantitative results are reported in the experimental section.**

***Index Terms*—Intrusion detection, Apache Spark, Feature selection, SHAP, Data leakage**

I. Introduction

Cyberattacks are growing in scale and sophistication, demanding IDS that are both accurate and scalable on large data [2], [3]. Machine learning has become the dominant approach for anomaly-based IDS, but modern network-flow datasets such as CICIDS2017 [1] provide nearly 80 features per flow, which increases training, inference, and memory cost — especially critical for edge deployment. Dimensionality reduction and feature selection are therefore key steps. However, different strategies such as embedded, unsupervised, or explanation-based ones measure "importance" in non-equivalent ways, and they are rarely compared under the same distributed pipeline.

A less-discussed but equally serious issue is the *trustworthiness of the evaluation pipeline*. Many reports on CICIDS2017 achieve near-perfect binary F1; this is usually a sign of data leakage rather than of model capability. *Data leakage* is the situation in which information that should not be available at real deployment time enters the training/evaluation process, inflating scores so that they no longer reflect true generalization. Labeling errors and duplicate flows in CICIDS2017 itself have been documented and shown to affect evaluation results substantially [4]. There are two common sources of leakage. First, the `destination_port` feature directly encodes the attacked service, because CICIDS2017 generates attack scenarios aimed at fixed ports, so it becomes a "shortcut" for the model to memorize the port-to-label mapping. Second, feature-level duplicate flows are not removed by full-duplicate removal, so a random split places the same sample into both the training and test sets. This paper handles both before comparing methods.

Main contributions. (1) A uniform comparison of RF Importance, SHAP, and PCA over the same eight Spark ML algorithms and two ensembles; (2) Integration of SHAP-based explanation [5] with a direct contrast against RF Importance [6]; (3) A *leakage-aware and statistically tested* evaluation pipeline: removing the leaking feature, deterministic feature-level deduplication (label-conflicting groups are

removed entirely), model selection on a validation set held out from the training set, and a paired permutation test (exact enumeration) over multiple resamplings with Nadeau–Bengio correction and effect size; (4) A per-attack-type multiclass evaluation, clarifying where the methods truly differ; (5) An *edge-aware* model-selection recommendation: jointly considering F1, the number of features, and model size for the Jetson Orin Nano Super.

Section II presents related work; Section III describes the method and evaluation pipeline; Section IV reports the experiments; Section V discusses; Section VI states threats to validity; and Section VII concludes.

II. Related Work

IDS surveys [2], [3], [7] show that both traditional machine learning and deep learning are widely applied on standard benchmarks such as CICIDS2017 [1]. Many works focus on a single algorithm, for instance online deep learning [8] or deep networks [9], without a uniform comparison of multiple dimensionality-reduction strategies on the same distributed pipeline. Apache Spark ML [10], [11] supports large-scale processing, but few studies combine feature selection, model explanation, and statistical testing in a single unified experimental framework. Recent edge IDS studies deploy deep-learning models on NVIDIA Jetson [12], [13], but they typically focus on a single model and on accuracy, and rarely compare multiple dimensionality-reduction strategies under a leakage-aware evaluation pipeline.

Feature selection. Variable-selection theory [14] distinguishes three groups: filter, wrapper, and embedded. Random Forest Importance [6] (embedded) and PCA [15] (unsupervised) are two popular directions for IDS but measure different quantities. SHAP [5] provides consistent local explanations and has been used for IDS [16], but is rarely contrasted directly with RF Importance on the same Spark training set.

Concept drift, anomaly, and evaluation. Concept drift [17] affects IDS stability over time; anomaly-detection studies [18]–[21] typically separate the filter from the classifier, and autoencoders are also used to reduce feature dimensionality for IDS on edge devices [21]. However, data leakage ("data snooping") and statistical-significance testing — identified as among the most common pitfalls of ML in security [22] — are rarely handled explicitly on CICIDS2017. This paper fills that gap. Table I summarizes the comparison.

III. Method

A. Overview of the Spark ML pipeline

The offline pipeline consists of sequential steps: data cleaning, deduplication, `VectorAssembler`, `StandardScaler`, an (optional) dimensionality-reduction transform, and then the classifier. Binary labels: Benign (0) and Attack (1). The entire `fit` of StandardScaler/PCA and the feature ranking use *only* the training set; the test set serves as the final holdout.

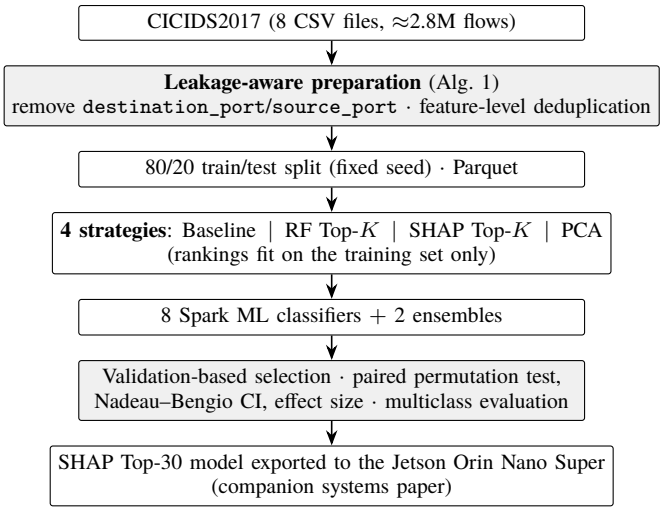

Figure 1. Leakage-aware evaluation pipeline: from raw CICIDS2017 flows to the statistically validated, edge-ready model.

B. Leakage-aware data preparation

We merge the eight CICIDS2017 CSV files, normalize column names, replace infinite/missing values, and then apply two leakage-removal steps *before* splitting the data:

- **Removing the label-leaking feature.** `destination_port` (and `source_port` if present) is removed because it directly encodes the target service of the attack; keeping it would let the model memorize the port-to-label mapping instead of learning flow behavior. An ablation (keep vs. remove the port) quantifies the impact.
- **Feature-level deduplication (deterministic).** Beyond removing fully duplicate records, we remove flows that are *identical across all feature columns* before splitting, preventing the same sample from appearing in both the training and test sets. For groups of flows that are feature-identical but have *conflicting labels* (Benign and Attack — a known error in CICIDS2017 [4]), the entire group is removed rather than arbitrarily keeping one row depending on data partitioning; the numbers of removed groups and rows are reported (270 groups, 44,260 rows), ensuring the deduplication result is deterministic and reproducible.

Algorithm 1 summarizes the procedure, and Figure 1 places it in the full evaluation pipeline. After processing, the data are split into **80% training / 20% test** (with a fixed seed), stored in Parquet format, and reused throughout the experiments. The number of remaining numeric features is 77, and the number of samples after deduplication is 1,785,349/447,275 (≈2,232,624 total after feature-level dedup).

C. Classifiers and the four dimensionality-reduction strategies

Eight algorithms: Decision Tree, Logistic Regression, SVM (LinearSVC), Naive Bayes, Random Forest, GBT, XGBoost, MLP; together with two ensembles (weighted Hybrid Bagging

Table I
CAPABILITY COMPARISON WITH RELATED IDS STUDIES ON CICIDS2017 AND SPARK/EDGE ("—": NOT ADDRESSED OR NOT REPORTED).

Reference	Dataset	Feature reduction / explainability	Leakage-aware	Statistical testing	Cross-dataset	Multi-class
Khraisat et al. [3]	Multiple (survey)	General	—	—	—	—
Vinayakumar et al. [9]	CICIDS2017	None	—	—	—	Yes
Ferrag et al. [7]	Multiple	None	—	—	—	Yes
Alrawashdeh & Purdy [8]	NSL-KDD	None	—	—	—	—
Sarhan et al. [16]	NetFlow (NF-*)	SHAP	—	—	Yes	—
Baalaji et al. [12]	IoT-edge	None	—	—	—	—
Asghar et al. [13]	Multiple	Metaheuristic	—	—	—	—
Hasan et al. [21]	IIoT	Autoencoder	—	—	—	—
The proposed method	**CICIDS2017, CSE-CIC-IDS2018**	**RF/SHAP/PCA**	**Yes**	**Yes**	**Yes**	**Yes**

Algorithm 1 Leakage-aware preparation of CICIDS2017

Input: CSV files $\{F_1, \ldots, F_N\}$; split ratio α=0.8
Output: D_{train}, D_{test} (Parquet)
$D \leftarrow \bigcup_i F_i$ ▷ normalize names; $\pm\infty \rightarrow$ NaN
$D \leftarrow \textsc{DropDuplicates}(D)$; $D \leftarrow \textsc{DropNaN}(D)$
$y_i \leftarrow 0$ if $\textsc{Upper}(\text{label}_i) = \text{BENIGN}$ else 1
$\mathcal{F} \leftarrow$ numeric columns $\setminus$ {`destination_port`, `source_port`, `protocol`, identifiers}
for all groups g of rows identical on all of $\mathcal{F}$ **do**
  if labels within g conflict (Benign and Attack) **then**
    drop g entirely ▷ contradictory ground truth; logged
  else keep one deterministic representative of g
split $D \rightarrow D_{\text{train}}/D_{\text{test}}$ (α, fixed seed); write Parquet

Top-3 and Soft Voting). LightGBM is excluded because its native library (via SynapseML) supports only x86_64 and cannot run on the Jetson Orin Nano Super edge device (ARM64), which is the deployment target of this study; therefore a LightGBM model cannot be included in the pipeline. Within the gradient-boosting family, XGBoost and GBT act as representatives. The four strategies: **Baseline** (all features); **RF Top-K** ($K \in \{20, 30, 40\}$); **SHAP Top-K**; **PCA** ($k \in \{20, 30, 40\}$). The RF and SHAP rankings use *only* the training set; SHAP is computed on a *class-stratified* sample so that both benign and attack traffic are represented. PCA is fixed at k=40, more than the Top-30 of the selection methods, because PCA is an *extraction*: each principal component is a linear combination carrying only part of the variance, so more components are needed to retain an equivalent amount of variance. Within the range $\{20, 30, 40\}$, k=40 retains the most variance and is therefore the fairest representative point for PCA.

D. Hyperparameters and class-imbalance handling

The hyperparameters (Table II) are *fixed a priori* and shared across all dimensionality-reduction methods, so that the observed differences correctly reflect the *feature-selection method* rather than per-model tuning; hyperparameter tuning is separated into a distinct stage (grid search with cross-validation) and applied only to the selected configuration. The stochastic models (RF, GBT, XGBoost, MLP) fix the `seed` for reproducibility. Class imbalance is handled *uniformly*. Every model that supports `weightCol` — namely Decision Tree, Logistic Regression, SVM, Naive Bayes, Random Forest, and GBT — is assigned class weights according to the benign/attack ratio, equivalent to XGBoost's `scale_pos_weight` mechanism. MLP alone does not support sample weights in Spark ML and is therefore trained without weights, as noted in the limitations.

E. Two-stage comparison design

To avoid selection bias from trying many configurations on the same test set, we separate the *exploration stage* (sweeping K, fully reported per K) from the *confirmation stage* (four configurations fixed a priori: Baseline, RF Top-30, SHAP Top-30, PCA k=40). The comparative conclusions rest on the four fixed configurations; we do not pick the configuration with the highest F1 after observing the results on the test set. Within each configuration, the *representative model* and the *method pair* entered into statistical testing are also chosen by F1 on a validation set (20% held out from the training set, fixed seed) — the test set does not participate in any selection step.

F. Rigorous evaluation pipeline

Metrics. Accuracy, Precision, Recall, F1, AUC-ROC, AUC-PR; training/prediction time; model size. Because the data are imbalanced, AUC-PR and attack-class recall are prioritized over accuracy.

In-distribution stability check. The representative model of each method is re-evaluated on a subset of the test set (disjoint from the training set). To be clear: this is only an *in-distribution stability* check, not a distribution-shift test; the true generalization assessment lies in the cross-dataset experiment (Section IV).

Statistical-significance testing. The two top methods are trained on N independent *resamplings* of the training/test split (default N=6), and in each round both use the *same* split so as to be paired. As a result, the variance reflects the effect of data sampling rather than only of model initialization. The 95% confidence interval is computed using the t distribution ($t_{0.975,N-1}$, not the value 1.96); because the resamplings overlap in training data, we report *in parallel* the Nadeau–Bengio [23] corrected confidence interval, with the variance multiplied by $(1/N + n_{\text{test}}/n_{\text{train}})$, and take the corrected interval as the basis for conclusions. Significance is tested

Table II
MAIN HYPERPARAMETERS OF THE EIGHT CLASSIFIERS (FIXED A PRIORI).

Model	Main hyperparameters
Decision Tree	maxDepth=15, minInstancesPerNode=5, impurity=entropy
Logistic Regression	maxIter=200, regParam=0.001, L2, threshold=0.5
SVM (LinearSVC)	maxIter=200, regParam=0.001
Naive Bayes	gaussian, smoothing=1.0
Random Forest	numTrees=200, maxDepth=15, featureSubset=sqrt
GBT	maxIter=150, maxDepth=6, stepSize=0.05, subsample=0.8
XGBoost	n_estimators=300, max_depth=8, lr=0.05, subsample=0.8, colsample=0.8
MLP	layers=[d,64,32,2], maxIter=150, stepSize=0.01
Class balancing: per-sample class weights (6 models) / positive-class weight scaling (XGBoost); MLP unweighted.	

by a *two-sided paired permutation test* on the per-round F1 difference, *exactly enumerating* all 2^N sign permutations (with N=6, that is 64 permutations) rather than Monte Carlo sampling; because the test is two-sided, the smallest attainable p is $2/2^N$, so $N \geq 6$ is needed to be able to reach below 0.05 (in contrast to a single-fixed-split / three-seed design, which cannot go below $p \approx 0.125$). Together with p, we report the paired Cohen's d effect size: with small N, a non-significant p-value is by itself weak evidence for "equivalence".

Multiclass evaluation. Because binary classification on CICIDS2017 is almost perfectly separable, we add a *per-attack-type multiclass* evaluation, comprising per-class precision/recall/F1, macro-F1, and a confusion matrix. It is here that rare classes such as Heartbleed, Infiltration, Web Attack, and Bot reveal the real differences among methods.

IV. EXPERIMENTS AND RESULTS

A. Setup

CICIDS2017; an 80/20 training/test split after leakage removal (Section III-B); Apache Spark 3.4.1, Java 17; hardware: a cluster of 2× Jetson Orin Nano Super (8 GB) and a coordinator host. Every script logs the run configuration (seed, environment variables, library versions) to ensure reproducibility. The full source code, configuration, and feature lists are publicly available at https://github.com/vuthainguyen1602/Thesis_IDS (branch `develop`); the pipeline can be reproduced on a single machine and does not require an edge-device cluster.

B. Port-feature ablation

Table III and Figure 2 quantify the impact of `destination_port`. After feature-level deduplication, the port's isolated effect is *small*: binary F1 is 0.9965 with the port and 0.9955 without it (a ≈0.001 gap) — the in-domain binary task is near saturation and the dominant leakage source (duplicate flows) is already handled; the direction (keeping the port yields slightly higher F1/recall) is nonetheless consistent with leakage. All subsequent results use the *port-removed* configuration.

Table III
IMPACT OF THE LEAKING FEATURE `DESTINATION_PORT` (RANDOM FOREST FIXED A PRIORI PER THE TABLE II CONFIGURATION, CLASS-WEIGHTED; BINARY).

Configuration	F1	Recall (Attack)	AUC-PR
Keep port	0.9965	0.9988	0.9999
Remove port (proposed)	0.9955	0.9961	0.9997

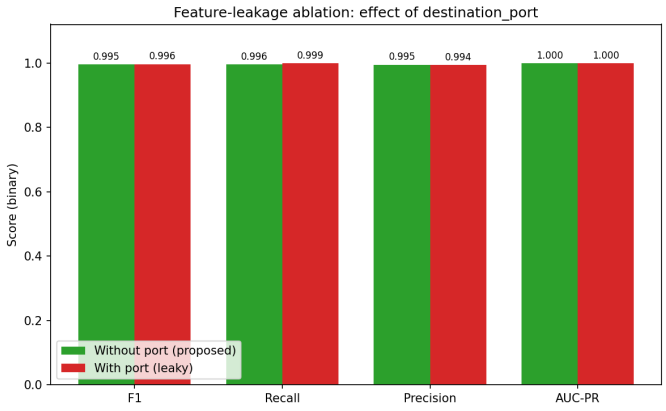

Figure 2. Impact of the leaking feature `destination_port` (Random Forest, binary).

C. Comparison of the four dimensionality-reduction methods

Table IV summarizes the best model per method, and Figure 3 visualizes F1 for each method × algorithm pair. The comparison design aims to verify whether feature selection (RF/SHAP Top-30) retains near-baseline performance while cutting ≈60% of the features; PCA k=40 reduces dimensionality in an unsupervised manner at the cost of explainability. All quantitative values are taken directly from the released pipeline outputs.

D. Statistical testing (paired resampling)

Table V reports the mean F1 ± 95% CI (t distribution) over N=6 resamplings for the two top methods, together with the paired permutation-test p-value. Conclusion: the two methods differ *significantly only at the floor* ($p \approx 0.031$, Cohen's $d \approx 1.94$), yet the absolute F1 gap is tiny (≈0.0001), so they are *practically equivalent*.

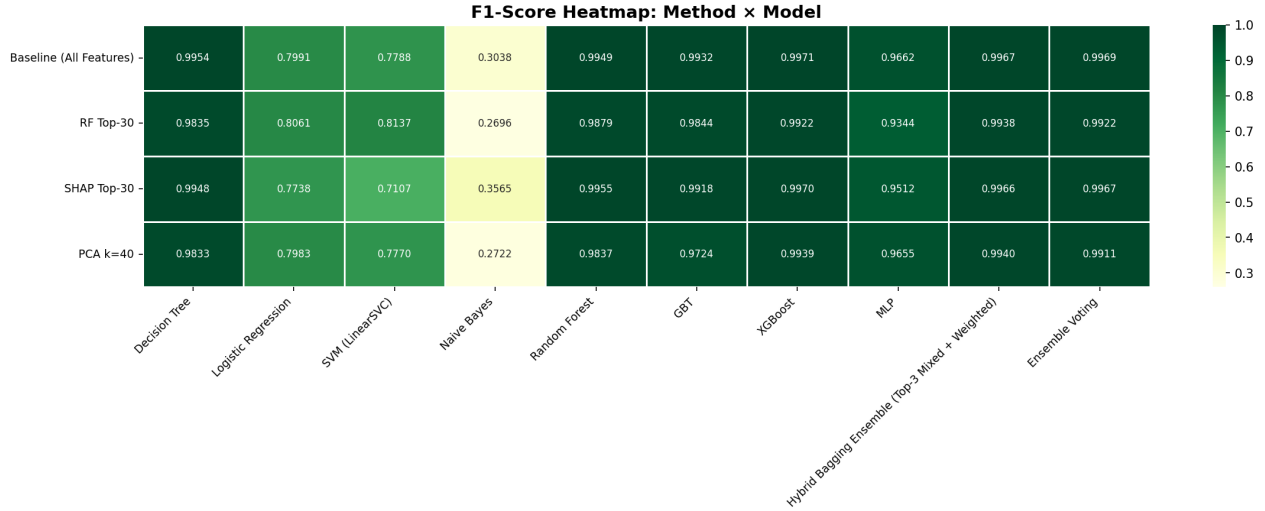

Figure 3. F1 heatmap by method and algorithm (binary, port removed).

Table IV
SUMMARY OF THE BEST MODEL PER METHOD (VALUES FROM THE RELEASED EXPERIMENT OUTPUTS).

Method	Model	F1	AUC-PR	Train (s)	Size (MB)
Baseline	XGBoost	0.9971	0.9999	4615.5	2.32
RF Top-30	XGBoost	0.9922	0.9998	1295.1	2.52
SHAP Top-30	XGBoost	0.9970	0.9999	1226.1	0.003†
PCA k=40	XGBoost	0.9939	0.9999	1444.8	3.82

†Sizes are obtained by serializing each fitted model *on the training cluster*; the distributed model-saving step writes the tree-ensemble data partitions on whichever executor holds them, so entries in the KB range (here 0.003 MB) captured only the driver-local metadata and *under-report* the true size. Since the size of a tree ensemble is governed by the number of trees and nodes rather than by the number of input features, the SHAP Top-30 XGBoost model is in reality comparable to its RF Top-30 counterpart (≈2.5 MB). For the same reason, a reduced feature set can even yield a slightly *larger* model than the baseline (2.52 vs. 2.32 MB): with fewer split candidates the trees need more nodes to reach comparable purity. The single-node export of the deployed pipeline measures model sizes reliably.

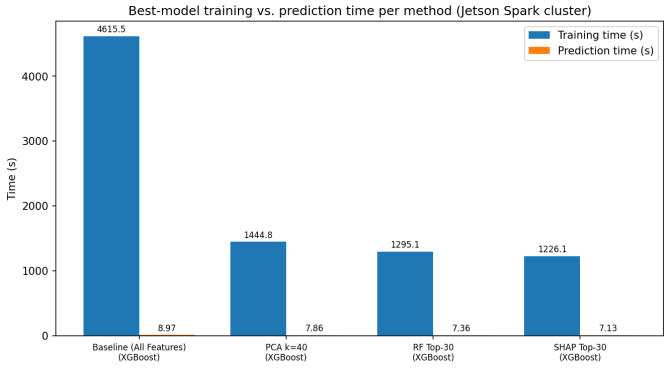

Figure 4. Training and inference time of the representative model of each method, measured on the Spark cluster used for training (Section IV); inference latency/throughput measurements on the edge device belong to a companion systems paper (under submission), not this paper.

Table V
MULTI-RESAMPLING STABILITY AND PAIRED TESTING (VALUES FROM THE RELEASED EXPERIMENT OUTPUTS).

Method	Model	F1 (mean ± CI95)	p (paired)
Baseline	XGBoost	0.99734 ± 0.00007	0.031
SHAP Top-30	XGBoost	0.99720 ± 0.00009	

E. Robustness and multiclass evaluation

In the in-distribution stability check (a subset of the test set), the ranking of the methods *is preserved* (XGBoost leads in all four methods); details in the released experiment outputs. The multiclass evaluation (Figure 5, Table VI) reveals a sharp contrast between weighted-F1 (≈0.998) and macro-F1 (≈0.806): the majority classes achieve high F1, while rare classes such as Web Attack (XSS, SQL Injection) and Bot have very low recall (0.03–0.46) — where feature selection makes the clearest difference. Reading the confusion matrix by *destination* of the errors separates two failure modes. XSS is mostly *mislabeled but still detected*: 81/111 samples are predicted as Web Brute Force, so at the binary (deployment) level 76% of XSS flows are still flagged as attacks. The dangerous mode is *leaking into Benign*: all 6 SQL Injection samples, 54% of Bot (158/290), and 27% of Web Brute Force are predicted benign. Three mechanisms drive this: extreme imbalance (6–311 samples vs. 380k benign); the removal of `destination_port`, which hits web attacks hardest —

without the port-80 shortcut their short HTTP flows resemble ordinary browsing and each other, so the honest 0.027 recall replaces a leakage-inflated one; and the intrinsic similarity of Bot C&C beaconing to benign background traffic once service identity is removed. This directly motivates targeted imbalance handling and temporal features for beaconing as future work.

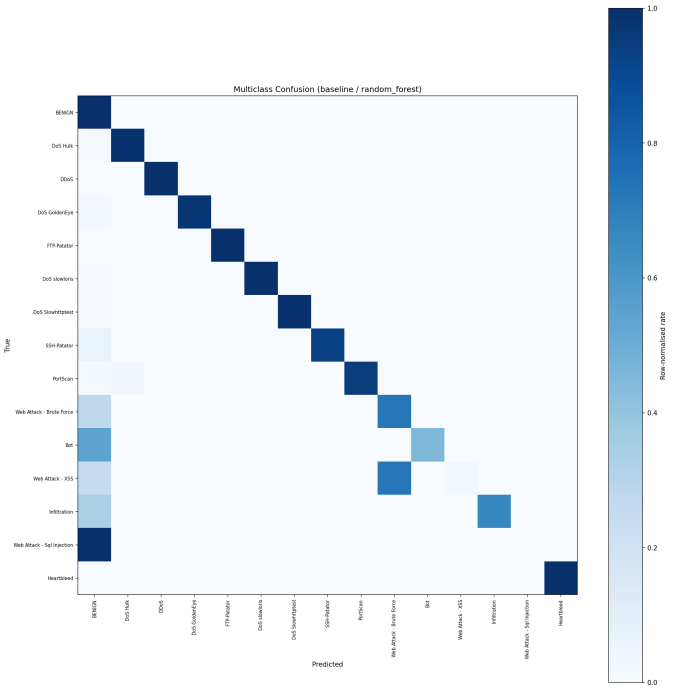

Figure 5. Multiclass confusion matrix (row-normalized).

Table VI
PER-CLASS PRECISION/RECALL/F1 (FROM THE RELEASED EXPERIMENT OUTPUTS).

Class	Support	Precision	Recall	F1
Benign	380,119	0.9981	0.9997	0.9989
DoS Hulk	34,446	0.9990	0.9904	0.9947
PortScan	383	1.0000	0.9530	0.9759
Web Attack – Brute Force	311	0.7346	0.7299	0.7323
Bot	290	1.0000	0.4552	0.6256
Web Attack – XSS	111	1.0000	0.0270	0.0526
Infiltration	12	1.0000	0.6667	0.8000
Web Attack – SQL Injection	6	0.0000	0.0000	0.0000
7 remaining classes, all F1 ≥ 0.96 — see the released per-class results				
macro-F1 $= 0.806$ weighted-F1 $= 0.998$				

F. Cross-dataset generalization

To go beyond *in-domain* evaluation, we train on one dataset and test on a *different* one (*cross-dataset* evaluation; and vice versa) on the *shared* leakage-free feature set of the two datasets. The gap between in-domain F1 (A→A) and cross-dataset F1 (A→B) in Table VII quantifies the *generalization* ability of the leakage-free model. This is a genuine distribution-shift test, unlike the in-distribution holdout above. Because CSE-CIC-IDS2018 [24] is very large (∼16 million flows), we use a *label-stratified* sample (fraction 0.2, ∼3.14M sampled flows, ∼2.39M after feature-level deduplication, fixed seed) that preserves the class proportions; CICIDS2017 uses ∼2.23M flows. The sample sizes are reported explicitly to ensure reproducibility.

Table VII
CROSS-DATASET GENERALIZATION (CICIDS2017 AND CSE-CIC-IDS2018, LABEL-STRATIFIED SAMPLE WITH A FIXED SEED; VALUES FROM THE RELEASED EXPERIMENT OUTPUTS).

Train	Test	Type	F1
CICIDS2017	CICIDS2017	in-domain	0.9952
CICIDS2017	CSE-CIC-2018	cross-dataset	0.0423
CSE-CIC-2018	CSE-CIC-2018	in-domain	0.9245
CSE-CIC-2018	CICIDS2017	cross-dataset	0.0535

The contrast is stark and, we argue, the most honest number in this paper: in-domain F1 is high in both directions (0.9952 and 0.9245), yet cross-dataset F1 collapses to 0.04–0.05, driven by near-zero recall (0.022 and 0.030) while precision in the 2017-to-2018 direction remains 0.60 — the model fails to *recognize* the other testbed's attacks rather than alarming indiscriminately. The shift has concrete causes: a different CICFlowMeter version (changed feature scales and definitions), a different network/service configuration, and non-overlapping attack tooling (e.g., HOIC/LOIC-HTTP exist only in 2018). This collapse is consistent with prior cross-dataset IDS studies [25], [26] and reinforces the central claim of this paper: near-perfect in-domain scores — even leakage-free ones — do not certify deployability under distribution shift, so cross-dataset evaluation must be a mandatory part of IDS validation, and domain adaptation (per-domain normalization, joint training, drift-aware updating) is a prerequisite for practical transfer.

V. DISCUSSION

Method comparison. For the tested pair (Baseline vs. SHAP Top-30, Table V), the permutation test is significant only at the floor of its resolution ($p \approx 0.031$) while the absolute F1 gap (≈0.0001) is far below the run-to-run variation; we therefore treat the two configurations as *practically equivalent*, and secondary criteria (explainability, size, stability, feature count) decide the deployment choice. RF Top-30, by contrast, trails SHAP Top-30 by ≈0.005 F1 at the same feature count. SHAP Top-30 provides consistent local explanations, an advantage for SOC operations.

Bridge to the edge device. Cutting ≈60% of the features directly reduces the vector-assembly cost and the model memory footprint on the Jetson Orin Nano Super. The selection criterion therefore needs to be *edge-aware*, that is, to consider not only F1 but also the number of features, the model size, and the training/prediction time (Figure 4). Latency and throughput measurements on the Jetson are presented in a companion systems paper (under submission).

Limitations of the binary evaluation. Removing the port and deduplicating brings binary F1 down to a realistic level, but binary classification remains a relatively easy task. Only

under multiclass evaluation do the weaknesses on rare attack classes emerge, and this is also a more reliable metric for distinguishing methods.

VI. Threats to Validity

Internal validity. The RF/SHAP rankings are derived from the original training set and kept fixed across resamplings. This is consistent with the "a priori configuration" design, but may introduce slight leakage in the ranking part. Re-ranking per resampling would be more rigorous but costly (especially SHAP); we make this trade-off explicit. The selection of the Top-3 component models for the ensemble is performed on a separate *validation set* held out from the training set, so the test set does not participate in model selection; the trade-off is that the base models are trained on a smaller portion of the data after extracting the validation set. In addition, class balancing is applied uniformly to seven models (Section III-D) but *not* to MLP, because Spark ML does not support sample weights for this model; the MLP result should therefore be read with that caveat.

External validity. CICIDS2017 (2017) may not reflect current traffic, and benchmark NIDS datasets carry documented quality limitations [27]; verification on newer datasets (CSE-CIC-IDS2018, CIC-IoT) is needed. The default random split shuffles time; a time-based split mode is supported for a more realistic evaluation. The cross-dataset generalization conclusion (Section IV-F) is based on a *representative stratified sample* of CSE-CIC-IDS2018 (due to edge resource limits), not the whole dataset; the sample size is reported explicitly, and scaling to the full dataset is the next direction. The model assumes an attack distribution approximating CICIDS2017 and has not been evaluated against deliberately evasive (adversarial evasion) traffic. Moreover, feature-level deduplication (Section III-B) itself changes the class distribution between benign and attack, which should be kept in mind when interpreting the results.

Statistical validity. The small sample size ($N{=}6$) gives the test low *power*: because the permutation test is two-sided, the smallest attainable p is only $2/2^N$, so $N{=}6$ just barely reaches the 0.05 threshold and can hardly detect small effects. Furthermore, the resamplings *overlap*, thus violating the independence assumption of the t distribution; therefore the Nadeau–Bengio corrected confidence interval is reported in parallel (Section III-F) and is the primary basis for conclusions, while the standard t interval is for reference only. A remaining limitation is that the RF/SHAP feature ranking is kept fixed across resamplings (no per-round re-ranking), creating a mild selection leakage favoring the two feature-selection branches; per-round re-ranking (feasible for RF) is the next reinforcement direction, together with increasing N with nested cross-validation.

VII. Conclusion

This paper compares four dimensionality-reduction strategies for IDS on Spark ML under a leakage-aware and statistically tested evaluation pipeline. The core contribution is not merely "which method is best" but also *how to evaluate reliably*: removing the leaking feature, deterministic feature-level deduplication, model selection on a validation set, exact-enumeration paired testing with Nadeau–Bengio correction, multiclass evaluation, and cross-dataset generalization between CICIDS2017 and CSE-CIC-IDS2018 (Section IV-F). Future directions: scaling to the full CSE-CIC-IDS2018 and to newer datasets (RoEduNet, CIC-IoT), verifying the transferability of the SHAP Top-30 set itself between the two datasets, hyperparameter optimization, and real-time deployment on the edge.

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